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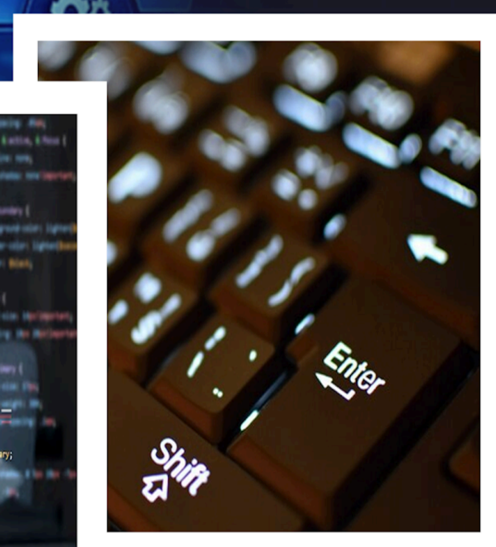
Computer Awareness

STUDY NOTES

Computer Software



Computer Software

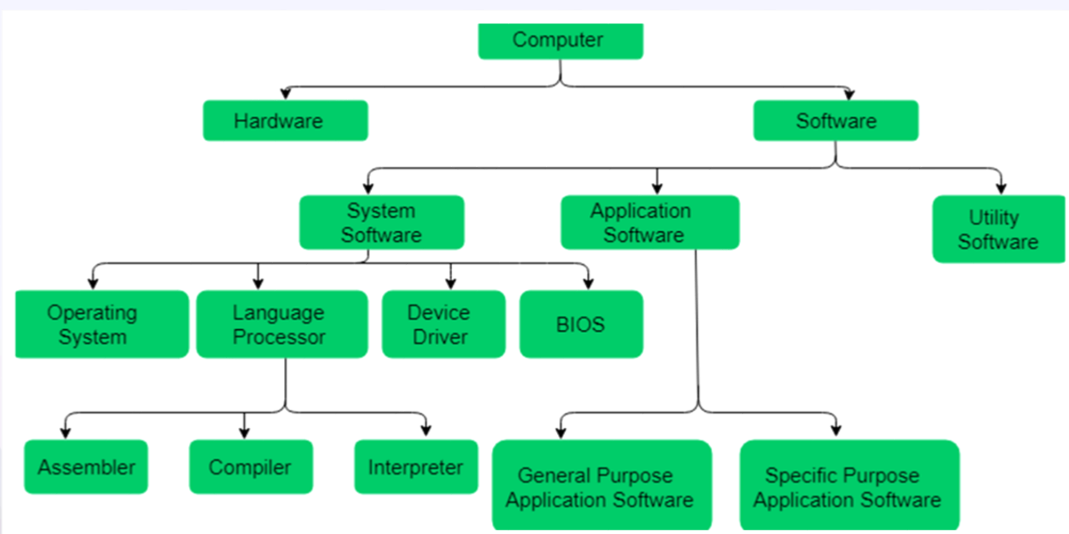
Software is a collection of instructions and data that tell a computer how to work. This is in contrast to physical hardware, from which the system is built and actually performs the work. In computer science and software engineering, computer software is all information processed by computer systems, including programs and data. Computer software includes computer programs, libraries and related non- executable data, such as online documentation or digital media. Computer hardware and software require each other and neither can be realistically used on its own.

A set of instructions that achieve a single outcome are called programs or procedures. Many programs function together to do a task to make a software.

For example, a word-processing software enables the user to create, edit and save documents. A web browser enables the user to view and share web pages and multimedia files. There are two categories of software –

- System Software
- Application Software
- Utility Software

Let us discuss them in detail.



System Software

Software required to run the hardware parts of the computer and other application software are called system software. System software acts as an interface between hardware and user applications. An interface is needed because hardware devices or machines and humans speak in different languages.

Machines understand only binary language i.e. 0 (absence of electric signal) and 1 (presence of electric signal) while humans speak in English, French, German, Tamil, Hindi and many other languages. English is the pre- dominant language for interacting with computers. Software is required to convert all human instructions into machine understandable instructions. And this is exactly what system software does.

Based on its function, system software is of three types –

- Operating System
- Language Processor
- Device Drivers

Operating System

System software that is responsible for functioning of all hardware parts and their interoperability to carry out tasks successfully is called **operating system (OS)**. OS is the first software to be loaded into computer memory when the computer is switched on and this is called **booting**. OS manages a computer's basic functions like storing data in memory, retrieving files from storage devices, scheduling tasks based on priority, etc.

10 Best Operating Systems In Market

1. MS-Windows
2. Ubuntu
3. Mac OS
4. Fedora
5. Solaris
6. Free BSD
7. Chrome OS
8. CentOS
9. Debian

10. Deepin

Language Processor

As discussed earlier, an important function of system software is to convert all user instructions into machine understandable language. When we talk of human machine interactions, languages are of three types –

- **Machine-level language** – This language is nothing but a string of 0s and 1s that the machines can understand. It is completely machine dependent.
- **Assembly-level language** – This language introduces a layer of abstraction by defining **mnemonics**. **Mnemonics** are English like words or symbols used to denote a long string of 0s and 1s. For example, the word "READ" can be defined to mean that a computer has to retrieve data from the memory. The complete **instruction** will also tell the memory address. Assembly level language is **machine dependent**.
- **High level language** – This language uses English like statements and is completely independent of machines. Programs written using high level languages are easy to create, read and understand.

Program written in high level programming languages like Java, C++, etc. is called **source code**. A set of instructions in machine readable form is called **object code** or **machine code**. **System software** that converts source code to object code is called **language processor**. There are three types of language interpreters–

- **Assembler** – Converts assembly level program into machine level program.
- **Interpreter** – Converts high level programs into machine level programs line by line.
- **Compiler** – Converts high level programs into machine level programs at one go rather than line by line.

Device Drivers

System software that controls and monitors functioning of a specific device on a computer is called **device driver**. Each device like printer, scanner, microphone, speaker, etc. that needs to be attached externally to the system has a specific driver associated with it. When you attach a new device, you need to install its driver so that the OS knows how it needs to be managed.

Many parts of a computer need drivers, and common examples are:

- Computer printers.
- Graphic cards.
- Modems.
- Network cards.
- Sound cards.

Application Software

- A software that performs a single task and nothing else is called application software. Application software is very specialized in its function and approach to solving a problem.
- Application programs interact with systems software; systems software then directs computer hardware to perform the necessary tasks.
- So a spreadsheet software can only do operations with numbers and nothing else. A hospital management software will manage hospital activities and nothing else. Here are some commonly used application software –
    - ◊ Word processing
    - ◊ Spreadsheet
    - ◊ Presentation
    - ◊ Database management
    - ◊ Multimedia tools

Utility Software

Application software that assists system software in doing their work is called **utility software**. Thus utility software is actually a cross between system software and application software. Examples of utility software include –

- Antivirus software
- Disk management tools
- File management tools
- Compression tools
- Backup tools

Middleware software

- Middleware is software that enables one or more kinds of communication or connectivity between two or more applications or application components in a distributed network.

- Common middleware examples include database middleware, application server middleware, message- oriented middleware, web middleware, and transaction-processing monitors.

Shareware software

Shareware software is software that is freely distributed to users on a trial basis. There is a time limit inbuilt in the software (for example- free for 30 days or 2 months). As the time limit gets over, it will be deactivated. To use it after the time limit, you have to pay for the software. Users prefer shareware because of following reasons –

- Available free of cost
- Helps to know about the product before buying it Some examples of freeware software are –
- Adobe acrobat 8 professional
- PHP Debugger 2.1.3.3
- Winzip
- Getright

Shareware are of following types:

- **Adware** – Contains ads to generate revenue for the developers
- **Donationware** – payment is optional
- **Nagware** – reminds user to purchase the license or the software
- **Freemium** – free for non-premium but of cost for premium features
- **Demoware** – demonstration version. It is further classified as crippleware and trialware.
- **Crippleware** – Some features are disabled under time-limit
- **Trialware** – all features are available under time-limit

Freeware software

Freeware software is a software that is *available free of cost*. An user can download freeware from the internet and use it. These softwares *do not provide any freedom of modifying, sharing and studying the program* as in open source software. Freeware is *closed source*. Users prefer freeware because of following reasons –

- Available free of cost
- Can be distributed free of cost

Some examples of freeware software are –

- **Adobe PDF**
- **yahoo messenger**
- **Google Talk**

- **MSN messenger**

Public domain software

Public domain software is not copyrighted. It is released without any conditions upon its use, and may be used without restriction. This type of software generally has the lowest level of support available.

Difference between freeware and shareware software

	Freeware	Shareware
Cost	Free of cost	Free for specific period of time later on asked the user to purchase.
Modify source code	Not allowed to cover all the copyrights.	Not allowed covered all the copyrights
Fully Functional	Usually	Certain features are provided for trial later on to unlock the rest features, users need to pay.
Features	All features are freely available	Some features are available for free trial or have limited use. To enable the rest of the features, payment should be made.
Distribution	Free of cost distributed among the people.	May or may not be freely distributed usually requires author's permission.
Access Duration time	Having no time limit	The program may only work for a short duration e.g. 30

		days.
Advantage	A free copyrighted software	Some features are free for trial and covered by copyright.
Disadvantage	Neither can sell these software nor can make any changes	It cannot be modified and it may be either a cut down or a temporary version.
Examples	MSN messenger, yahoo messenger, Adobe PDF,etc.	Winzip,CuteFTP, Getright,etc.

Open Source software

Open Source software is the software that is available to users with source code. Source code is a part of a program or software. Users can modify, inspect and enhance it to improve the software. Additional features can be added in the source code. Users use source code to copy, learn and share it. An Open source software can either be free of cost or chargeable. Users prefer open source software because of following reasons-

- More control over the software
- More secure
- Stable
- High quality results
- Helps in becoming a better programmer as you can learn and develop from the source code to make new softwares.

Examples of open source software are –

- Apache HTTP web server
- Mozilla’s Firefox web browser
- Thunderbird email client
- Database system
- GNU compiler collection

- Moodle
- OpenOffice
- PHP
- Perl

Proprietary Software

Proprietary software refers to the software which is owned by an individual or a company. There are restrictions on its distribution and use. It is also sometimes known as closed-source or commercial software.

Some of the advantages associated with proprietary software are –

- Stability – Its releases are stable throughout and the software does not crash easily.
- Reliable – It can be relied upon during handling of critical processes.
- Uniqueness – It is unique in nature and does not have many alternatives.
- Compatibility – It is compatible with many Operating systems.

Some of its disadvantages are –

- It is costly in nature.
- Source code is not accessible for modification. Redistribution is forbidden.

Examples of Proprietary Software –

- MacOS
- Microsoft Windows Professional Edition
- Adobe Suite